MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 1102: INTRODUCTION TO PROGRAMMING WITH C

SUPPLEMENTARY EXAMINATION

DATE: DECEMBER 2021 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. There are **15** questions on this paper. **ANSWER ALL.**

i. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

1. Define the following terms as used in programming:

   a. Variable. **[1 Mark]**
   b. Compiler. **[1 Mark]**
   c. Function. **[1 Mark]**
   d. Conditional loop. **[1 Mark]**
   e. Assignment. **[1 Mark]**

2. Write an algorithm in a form of flow chart for a program that asks a user to enter a whole number. If the number is even, the program should calculate and print the cube of that number otherwise it should calculate and print the square of that number. **[4 Marks]**

3. What will be the output from the following program segment **(There are no errors)**: **[4 Marks]**

```
int x=4,y=2;
printf("The values are: ");
printf("%d ", ++x);
printf("%d ", x + --y);
```

4. Write a program to calculate and print the salary of a new recruit as follows; if salary is below K600, tax should not be calculated otherwise a 0.2% tax should be deducted. The program should reject salary amount that is less than zero. **[5 Marks]**

5. Write a program that expects a certain number from a user (e.g 12). The user should keep entering the numbers until the intended number has been entered. When a user enters a number that is not required, the program should print a message "**Mwalakwitsa. Yesaniso**", and allow the user to enter another number. Make an assumption on which number to use. **[5 Marks]**

6. A user wants to implement a function that will calculate bonus of a salesman who will get 10% of the total sales. Write a function prototype that will achieve this. Let the name of the function be ***BonusCalculator***. **[4 Marks]**

7. What is main the difference between a while loop and a do while loop? **[2 Marks]**

8. Write a programme that calculates and prints the average of three decimal numbers entered by a user. **[4 Marks]**

9. What do you understand by the term infinite loop? Write a simple program that implements such type of a loop. **[4 Marks]**

10. Explain how a for loop work. **[3 Marks]**

11. Use a for loop to write a program that calculates and prints a multiplication table as follows: *(Hint: Use nested for loops)*

2	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**
3	**6**	**9**	**12**	**15**	**18**	**21**	**24**	**27**
4	**8**	**12**	**16**	**20**	**24**	**28**	**32**	**36**

[8 Marks]

12. Explain the **two** arguments of a scanf function. **[4 Marks]**

13. Determine and correct the syntax errors from the following statements:

a. char letter = 2; **[2 Marks]**
b. float int; **[2 Marks]**
c. int X= 8; printf("%d", x); **[2 Marks]**
d. int y; for (y=0; y=10, --y) {printf("%d\n", y*y)} **[2 Marks]**
e. float tax = = 20%; **[2 Marks]**
f. char number = "21"; **[2 Marks]**

14. What is the difference between a prefix increment and post-fix increment? **[2 Marks]**

END OF QUESTION PAPER